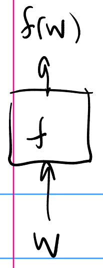
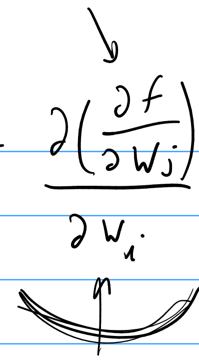
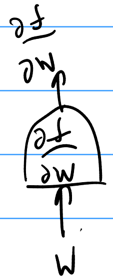


$$\frac{\partial f}{\partial w} = [\ N\ ]$$

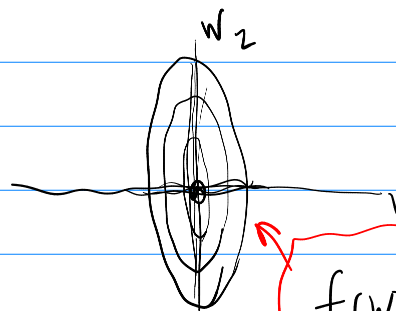
$$\frac{\partial^2 f(w)}{\partial w^2} = \begin{bmatrix} & N \\ N & \end{bmatrix}$$

$$\left(\frac{\partial^2 f(w)}{\partial w^2}\right)_{ij} = \frac{\partial^2 f(w)}{\partial w_i \partial w_j} = \frac{\partial\left(\frac{\partial f}{\partial w_j}\right)}{\partial w_i}$$





$$H = \frac{\partial^2 f(w)}{\partial w^2} = \begin{array}{c c} & j \\ i & \begin{bmatrix} \frac{\partial^2 f}{\partial w_1 \partial w_1} & & \\ & \frac{\partial^2 f(w)}{\partial w_i \partial w_j} & \\ & & \ddots \end{bmatrix} \end{array}$$

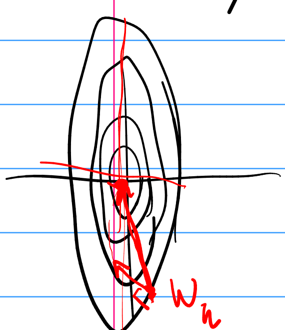


$$\frac{\partial^2 f}{\partial w_1^2} = 2 \qquad \frac{\partial^2 f}{\partial w_2^2} = 1$$

$$\boxed{f(w) = \frac{2}{2} w_1^2 + \frac{1}{2} w_2^2}$$

$$\frac{\partial f}{\partial w_1} = 2w_1 \qquad \frac{\partial^2 f}{\partial w_1^2} = 2$$

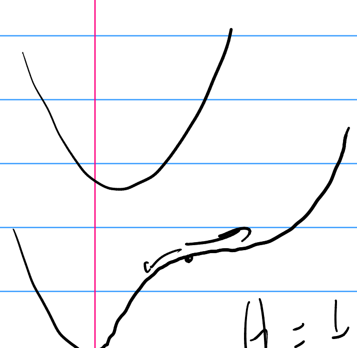
$$\frac{\partial^2 f}{\partial w_1 \partial w_2} = \frac{\partial(2w_1)}{\partial w_2} = 0 \qquad H = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\tilde{w} = w_k - H^{-1} \frac{\partial f}{\partial w}^T = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{bmatrix} \frac{\partial f}{\partial w}^T \qquad \frac{\partial f}{\partial w} = 0$$

Newton's Algorithms

$$g(w) = 0$$



Levenberg Marquardt method

$$\tilde{H} \qquad f(w) = \frac{1}{p} \sum_{i=1}^{p} L(\ w) = \frac{1}{2} \| y - g(z,w) \|^2$$

$$H = \frac{1}{2p} \sum_{i=1}^{p} \frac{\partial \left(\frac{\partial \| y_i - g(z_i,w) \|^2}{\partial w} \right)}{\partial w} = \frac{1}{2p} \sum_i \frac{\partial \left[2(y_i - g(z_i,w)) \frac{\partial g(z_i,w)}{\partial w} \right]}{\partial w}$$

$$H = \frac{\partial}{\partial w}\left[\frac{1}{2P}\sum_i \left(y_i - g(x_i, w)\right)\frac{\partial g(x_i,w)}{\partial w}\right] =$$

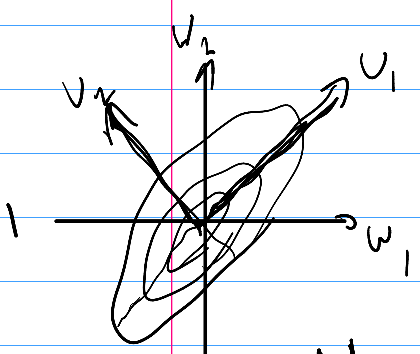
3th order tensor

$$\frac{1}{2p}\sum_i \frac{\partial g(x_i, w)}{\partial w}^T \frac{\partial g(x,w)}{\partial w} + \underbrace{\left(y_i - g(x_i,w)\right)^T \frac{\partial^2 g(x_i,w)}{\partial w^2}}_{\approx 0}$$

Gauss-Newton approx $\tilde{H} = \frac{1}{2p}\sum_i \frac{\partial g(x_i,w)}{\partial w}^T \frac{\partial g(x_i,w)}{\partial w}$

Levenberg Marquardt $\tilde{H} = \frac{1}{2p}\sum_i \frac{\partial g(x_i,w)}{\partial w}^T \frac{\partial g(x_i,w)}{\partial w} + \varepsilon I$

$\tilde{H}_{ii}$



$$f(u) = \frac{2}{2}u_1^2 + \frac{1}{2}u_2^2 = \frac{1}{2}u^T\begin{bmatrix}2 & 0\\ 0 & 1\end{bmatrix}u$$

$$f(w) = \frac{1}{2}w^T Q^T \begin{bmatrix}2 & 0\\ 0 & 1\end{bmatrix} Q w$$

$u = Qw$

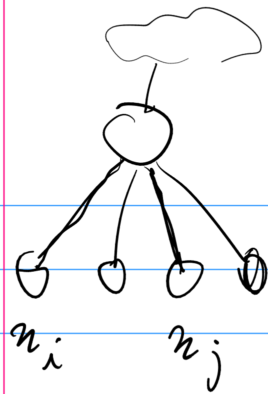
$$f(w) = \frac{1}{2}w^T \underbrace{\left[Q^T\begin{bmatrix}2 & 0\\ 0 & 1\end{bmatrix}Q\right]}_{H} w$$

$$f(w) = \frac{1}{2}w^T H w$$

$$f(w) = \frac{1}{2P}\sum_i \left(y_i - w^T x_i\right)^2$$

$$\frac{\partial f}{\partial w} = \frac{1}{p}\sum_i \left(y_i - w^T x_i\right) x_i^T$$

$$\boxed{H = \frac{\partial^2 f}{\partial w \partial w^T} = \frac{1}{p}\sum_i x_i x_i^T}$$



H ideally $= I$

$$H_{ij} = \frac{1}{p}\sum_{k=1}^{p} x_{ik}\, x_{jk} = 0 \text{ ideally}$$

$$H_{ii} = \frac{1}{p}\sum_{k=1}^{p} x_{ik}^{2} = 1 \text{ ideally}$$

#1 variables should have zero mean

#2 variables should have variance 1 (std dev = 1)